

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2022-23

Paper No. BPT-505: Electromagnetic Theory

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer all questions. Symbols have their usual meanings.***Question 1.** Answer any **four** of the following:

[4 × 5 = 20]

(a) For the vector field:

Question 2.

- (i) In a lossy dielectric medium, show that $\tan \delta = \epsilon''/\epsilon'$, where δ is the loss angle of the medium. Show how the phase of the intrinsic impedance is related to the loss angle. [8]
- (ii) Consider the multipole expansion of the electrostatic potential and show that the dipole term can be written in the form:

Question 3.

- (iii) Consider Ampere's law in the presence of magnetic materials. Write it in both differential and integral forms. Obtain the magnetostatic boundary conditions using the integral form of the equations. [8]
- (iv) Discuss the propagation of electromagnetic waves in a lossy dielectric medium and derive the expression for the intrinsic impedance of the medium. [6]

OR

- (v) What do you mean by the divergence of a vector field? Calculate the divergence of:

Question 4. Obtain the electrostatic potential both inside and outside of a hollow sphere of radius R which is maintained at a constant surface charge density σ_0 . [12]

Question 5.

- (vi) Show that the TE_{10} mode is the dominant mode (lowest cutoff frequency) of all the modes supported by a rectangular waveguide. [8]
- (vii) Derive the relation between reflection and transmission coefficients for an obliquely incident electromagnetic wave at a dielectric interface where the wave electric field is parallel to the plane of incidence (parallel polarization). [6]

OR

- (viii) A plane electromagnetic wave is normally incident on the boundary $z = 0$ formed by a lossless dielectric medium and a good conductor. Obtain the z -values at which the minimum value of $|E_{1y}|$ occurs in medium 1, where E_{1y} is the total electric field in medium 1. [8]
- (ix) Show that a TE_{10} mode of a rectangular waveguide can be represented as a sum of two plane TEM waves propagating along zigzag paths between the guide walls. [6]